Crystal Meet Rooms: architecture

An internal, device-specific Zoom Meeting SDK app. Every component is on Crystal PM's office network except Zoom itself.

CRYSTAL PM OFFICE NETWORK (LAN ONLY, NO INBOUND INTERNET ACCESS)

Table touch screen

Kiosk browser showing the room page served by the room device.

Join now, Mute, Camera, Leave, paste a link. A person presses Join; nothing joins by itself.

:8080

Door sign

PoE panel showing free / booked / in use, read from the room device.

:8080/sign

Dashboard (Raspberry Pi)

Node/Express + Postgres under Docker. Fleet status and enrollment tokens.

Never talks to Zoom.

WS :3001

Room device: Raspberry Pi 5 behind the room's TV (one per room)

Raspberry Pi OS 64-bit. The croom agent, Python 3.12, runs as a systemd service.

Control service

aiohttp on :8080, LAN only. Room page, door sign, /api/meeting/*.

Calendar service

Google Calendar read-only via a service account (today's bookings).

Meeting service, Zoom provider

1. Mints the Meeting SDK signature (HS256 JWT) from the SDK app's client id and secret.
2. For a meeting hosted by another account: Server-to-Server OAuth token, then GET /users/<room user>/token?type=zak.
3. Serves a one-page site on 127.0.0.1 that loads Zoom's Web Meeting SDK 6.5.0 (Client View) from source.zoom.us.
4. Opens it in the bundled Chromium (Playwright) on the TV; mute, camera and leave go through the SDK's bridge.

Credentials

/etc/croom/zoom-credentials.json (service user, mode 600): both apps' client ids and secrets, the account id, the room's Zoom user. Nothing else is stored: the ZAK lives in memory for one join, and the join page is sent with no-store, so the browser keeps no copy.

Camera and speakerphone are USB devices on the room device; media goes straight from Chromium to Zoom's meeting servers over TLS/DTLS-SRTP, as in any Zoom web client.

token (HTTPS)

ZAK (HTTPS)

SDK scripts (HTTPS)

join + media

ZOOM

App Marketplace

General App "Crystal Meet Rooms" (Meeting SDK) and a Server-to-Server OAuth app with scope user:read:zak:admin. Unlisted.

zoom.us/oauth/token

account_credentials grant (HTTPS). Token cached in memory until expiry.

api.zoom.us/v2/users/{user}/token

type=zak for the room's own Zoom user (HTTPS).

source.zoom.us

Zoom's Web Meeting SDK scripts and styles (HTTPS).

Zoom meeting servers

The room joins as a participant: signature + meeting number + passcode (+ ZAK for other accounts' meetings).